

Rab Nawaz, Ph.D.

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SUMMARY

Lecturer and Technical Trainer in Data & AI with experience designing and delivering high-impact learning programs across machine learning, Python, and applied AI. Skilled in translating complex concepts into engaging, hands-on training for technical and non-technical audiences. Experienced in generative AI, prompt engineering, and real-world ML applications, with exposure to cloud-based AI services (AWS). Strong focus on customer adoption, capability building, and practical implementation of AI solutions.

PROFESSIONAL EXPERIENCE

Lecturer & Technical Trainer – Data & AI

2023 – Present

University of Essex, Colchester

England, UK

Design and deliver applied AI and data science training through lectures, workshops, and hands-on labs, focusing on practical implementation and real-world problem solving.

- Deliver instructor-led training in data analysis, machine learning, and applied AI.
- Integrate generative AI concepts including LLMs and prompt engineering into training materials.
- Design structured learning materials linking AI concepts to real-world use cases.
- Develop interactive tools and demos for technical and non-technical users.
- Incorporate real-world tools such as Jupyter, APIs, and deployed ML systems into teaching.
- Adapt training delivery based on learner feedback and engagement.

Consultant & Programme Lead – Data Science

06/2025 – 04/2026

FutureWorks Training Limited, Manchester

England, UK

Designed a full data science apprenticeship programme aligned with industry standards and business needs.

- Designed modular training covering machine learning, AI, data engineering, and deployment.
- Incorporated generative AI, LLM applications, and prompt engineering into programme content.
- Translated complex AI concepts into scenario-based learning for professional learners.
- Structured real-world projects simulating enterprise AI implementation.
- Embedded responsible AI, governance, and GDPR considerations.

Data & AI Specialist

10/2021 – 11/2023

University of Glasgow, Glasgow

Scotland, UK

Delivered end-to-end data science solutions across clinical and industry projects.

- Developed machine learning models using Python (NumPy, Pandas, Scikit-learn).
- Built reproducible analytical pipelines for real-world applications.
- Delivered training and knowledge transfer sessions to stakeholders.
- Translated analytical outputs into clear, actionable insights.
- Built user-facing tools supporting adoption of AI workflows.

Visiting Researcher – Data & AI Applications

2020 – 2021

University of Graz

Austria

- Delivered analytical workflows using real-world datasets.
- Supported knowledge transfer through documentation and visualisation.

PROJECTS

Loan Default Risk Prediction – AI System

2025

Built an end-to-end machine learning system for loan default prediction using Python, PyTorch, scikit-learn and XGBoost. Designed modular pipelines and deployed via FastAPI with Docker and CI/CD workflows.

- Designed data ingestion, cleaning, and feature engineering pipelines.

- Trained and evaluated multiple models including neural networks.
- Built API-based inference system using FastAPI.
- Implemented CI/CD pipelines using GitHub Actions.
- Deployed models using HuggingFace for scalable inference.

Cognitive Workload Classification – AI System

2024

Developed a machine learning system for cognitive workload classification using time-series data.

- Built full data processing and feature engineering pipeline.
- Implemented SVM models with cross-validation.
- Designed modular training and inference workflows.
- Developed a GUI enabling non-technical users to interact with AI models.

SKILLS

Training & Enablement

- Customer-facing technical training
- AI enablement programmes
- Workshops and labs
- Curriculum design
- Mentoring and knowledge transfer

Programming & Tools

- Python (NumPy, Pandas, Scikit-learn)
- PyTorch
- HuggingFace
- FastAPI, Docker, GitHub Actions
- Jupyter Notebook, VS Code

AI & Machine Learning

- Machine learning and statistical modelling
- Generative AI and LLM fundamentals
- Prompt engineering
- Model evaluation and interpretation

Platforms & Cloud

- AWS AI services (Bedrock, SageMaker)
- Understanding of cloud AI workflows
- Starburst Fundamentals

AWARDS & ACHIEVEMENTS

Global Talent award for Exceptional Research Promise <i>Royal Society of Engineering, UK</i>	2023
Best Paper Award <i>International Conference on Emerging Trends in Smart Technologies</i>	2022
Best Paper Award <i>International Conference on BioSignal Analysis, Processing and Systems</i>	2018
PhD Studentship Award <i>Universiti Tunku Abdul Rahman, Malaysia</i>	2017
Excellent Trainee Engineer Award <i>ZTE Telecom, Pakistan</i>	2015

EDUCATION

PhD in Engineering Universiti Tunku Abdul Rahman (UTAR), Malaysia	2021
MSc in Electrical Engineering National University of Sciences and Technology (NUST), Pakistan	2016
BSc in Electronics Engineering International Islamic University Islamabad (IIUI), Pakistan	2013